DSP Assignment - Emotion Recognition from EEG Brain Signals using SVM

Vibha Rao
Roll No: 241EC162

Vikram Prem Kumar
Roll No: 241EC163

April 7, 2026

The source code for implementation and visualizations can be found at:
`Link To Github Repo`

The original paper is available at:
`Emotion Recognition from EEG-brain Signals`

1 Introduction

1.1 Abstract

In this work, we implement the paper titled "Human Emotion Recognition from EEG-brain Signals using Enhanced Machine Learning Method". The study explores the use of EEG data for detecting emotions with help of Machine Learning Models. Using the `SEED IV` dataset we apply a pipeline consisting of signal preprocessing, time and frequency domain feature extraction, and an attention-based feature selection mechanism (AAFST) to classify EEG signals into four emotional states: happiness, sadness, fear, and neutrality. Classification is performed using a multi-class Support Vector Machine (SVM). Our implementation successfully replicates the core methodology of the original paper, achieving a classification accuracy of 85%.

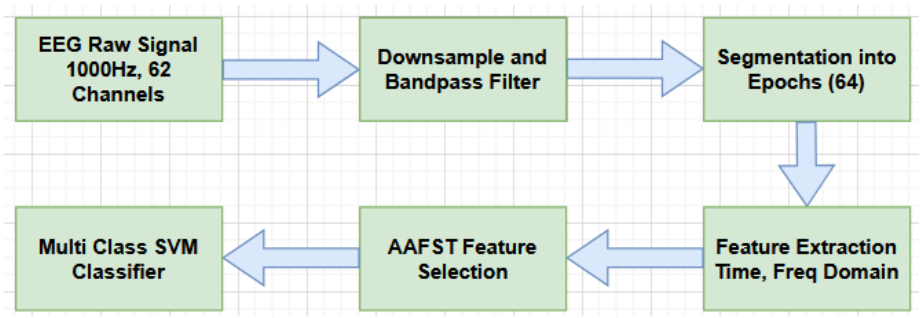

Figure 1: Flowchart description of System

1.2 Electroencephalography (EEG)

Electroencephalography (EEG) is a non-invasive neurophysiological technique that measures brain activity by recording voltage fluctuations caused by ionic currents within neurons. First developed by Hans Berger in 1924, EEG is widely used in clinical and research settings due to its relatively low cost compared to other neuroimaging methods such as fMRI or PET.

EEG signals are collected using electrodes placed on the scalp, typically following standardized systems like the International 10–20 system. Each electrode captures the combined activity of thousands of neurons. The signals are low amplitude (in microvolts, μV) and highly susceptible to artifacts from muscle movement, eye blinks, and external interference. Therefore, preprocessing steps such as filtering and downsampling are essential to extract meaningful neural information. EEG signals can be decomposed into distinct frequency bands, each associated with specific cognitive and emotional states:

- **Delta (0.5–4 Hz):** Predominantly observed during deep sleep and unconscious states. Elevated delta activity has also been linked to certain neurological conditions and states of reduced arousal.
- **Theta (4–8 Hz):** Associated with drowsiness, light sleep, meditation, and memory encoding. Theta waves are often observed during emotional processing and have been linked to states of stress and anxiety.
- **Alpha (8–13 Hz):** Typically present during relaxed, wakeful states with eyes closed. Alpha suppression, or a decrease in alpha power, is commonly observed during active cognitive processing and heightened emotional arousal.
- **Beta (13–30 Hz):** Dominant during active thinking, focused attention, and states of alertness. High beta activity is often associated with anxiety, excitement, and cognitive engagement.
- **Gamma (30–75 Hz):** Linked to higher-order cognitive functions such as perception, attention, and conscious awareness. Gamma oscillations are thought to reflect the binding of information across distributed brain regions.

In Figure 2 we have plotted a single channel of the raw eeg signal and decomposed into the 5 frequency bands mentioned above.

1.3 Machine Learning for EEG-Based Emotion Recognition

The paper follow a traditional ML approach - starting from preprocessing followed by feature extraction and classification. Handcrafted features derived from time-domain statistics, frequency-domain power estimates, and time-frequency representations are computed from segmented EEG epochs and fed into a SVM classifier.

1.4 Dataset Description

The SEED-IV dataset is a publicly available EEG dataset. The dataset comprises EEG recordings from 15 participants, each completing 3 sessions on separate days, resulting in a total of 45 recording

sessions. During each session, participants watched 24 carefully selected video clips and their emotions were recorded, happiness, sadness, fear, and neutrality.

EEG signals were recorded at a sampling rate of 1000 Hz using a 62-channel electrode cap following the international 10-20 system. The raw EEG signals are stored in `.mat` format, with each file containing 24 trials of shape $62 \times S$, where 62 represents the number of electrodes and S represents the total number of samples recorded during the trial.

Figure 2: Raw EEG decomposed into 5 frequency bands (Delta to Gamma)

2 Methodology

2.1 Preprocessing

We follow the preprocessing and data preparation strategies as given in the paper. Raw EEG signals undergo a series of preprocessing steps to enhance signal quality and remove noise prior to feature extraction. First, the signals are downsampled from the original sampling rate of 1000 Hz to 200 Hz, reducing computational overhead while preserving relevant frequency information. It is then followed by filtering, a fourth-order Butterworth bandpass filter is applied to each channel, retaining frequencies in the range of 0.5–75 Hz. This eliminates low-frequency artifacts, as well as high-frequency noise. The preprocessed signal retains the shape $62 \times S'$, where S' denotes the number of samples after downsampling.

Following filtering, each trial is segmented into fixed-length epochs of 800 samples, corresponding to 4 seconds of EEG activity at 200 Hz. To maximize the number of training samples available for classification, a sliding window approach is employed, generating 64 overlapping epochs per trial. Each epoch inherits the emotion label of its parent trial, resulting in a total of $15 \times 3 \times 24 \times 64 = 69,120$ epochs throughout the data set.

Figure 3 depicts the effect of preprocessing via downsampling and filtering of a single channel of raw EEG signal.

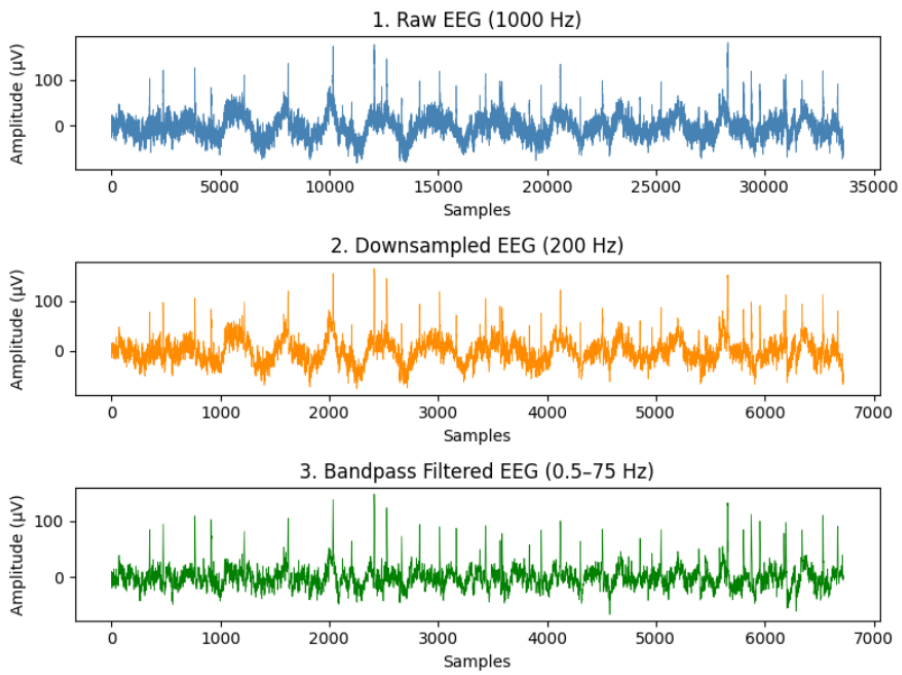

Figure 3: Visualization of Raw, Downsampled (200Hz) and Bandpass Filtered EEG Signal

2.2 Feature Extraction

For each epoch of shape 62×800, a set of time-domain and frequency-domain features are extracted independently from each of the 62 channels, yielding a comprehensive feature representation of the underlying neural activity.

2.2.1 Time-Domain Features

Time-domain features capture the statistical characteristics of the raw EEG signal and reflect the dynamics of neural activity over the epoch window.

Mean measures the average amplitude of the signal and reflects the baseline neural activity level:

$$\mu = \frac{1}{N} \sum_{n=1}^{N} x_n \tag{1}$$

Standard Deviation measures the dispersion of the signal amplitude around the mean, providing a direct indication of signal variability:

$$\sigma = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2} \tag{2}$$

Variance quantifies the spread of the signal around its mean, reflecting the overall power of neural oscillations:

$$\sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2 \tag{3}$$

Skewness measures the asymmetry of the signal distribution, capturing irregularities in neural firing patterns:

$$S = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{x_n - \mu}{\sigma}\right)^3 \tag{4}$$

Kurtosis measures the peakedness of the signal distribution, and is particularly sensitive to transient high-amplitude events such as spike activity:

$$K = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{x_n - \mu}{\sigma}\right)^4 \tag{5}$$

Hjorth Mobility characterizes the mean frequency of the signal by computing the ratio of the standard deviation of the first derivative to the standard deviation of the signal:

$$M = \sqrt{\frac{\mathrm{Var}(x')}{\mathrm{Var}(x)}} \tag{6}$$

where x' denotes the first-order difference of the signal x.

2.2.2 Frequency-Domain Features

Frequency-domain features characterize the spectral content of the EEG signal across five physiologically meaningful frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–75 Hz).

Power Spectral Density (PSD) is estimated using Welch's method, which averages periodograms computed over overlapping signal segments to produce a stable spectral estimate. The mean PSD within each frequency band $[f_1, f_2]$ is computed as:

$$\mathrm{PSD}_{\mathrm{band}} = \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}} P(f) \tag{7}$$

where $\mathcal{F}$ denotes the set of frequency bins within the band and $P(f)$ is the power at frequency f.

Band Power quantifies the total spectral power within each frequency band via numerical integration using Simpson's rule:

$$\mathrm{BP}_{\mathrm{band}} = \int_{f_1}^{f_2} P(f)\,df \tag{8}$$

Band power is computed for the delta, theta, alpha, and beta bands, capturing the energy distribution across the primary neural rhythms.

Differential Entropy (DE) is a measure of the complexity of a continuous-valued signal. For EEG signals, it is approximated as the logarithm of the variance of the bandpass-filtered signal within each frequency band:

$$\mathrm{DE} = -\int P(x) \ln P(x)\, dx \tag{9}$$

where x_{band} is the EEG signal filtered to the corresponding frequency band.

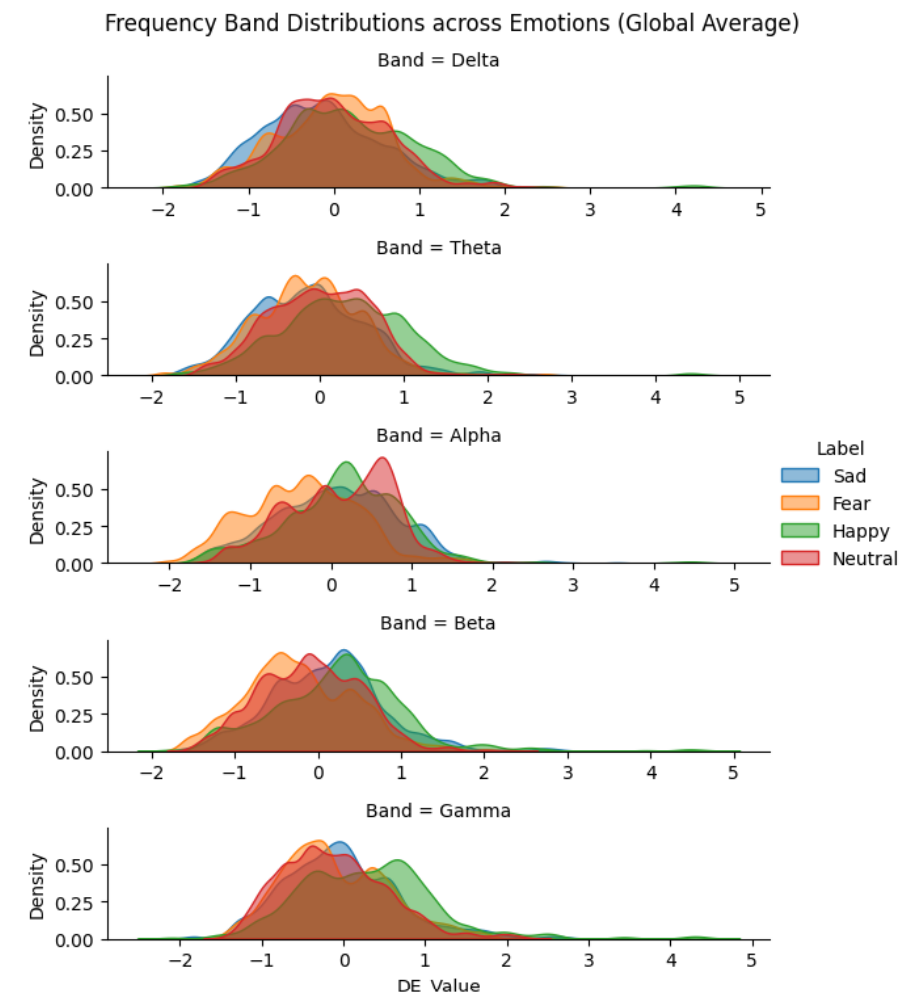

Figure 4: Band Power Distributions

2.3 Adaptive attention-based feature selection and transformation (AAFST)

Following feature extraction, the high-dimensional feature vectors are passed through an Adaptive Attention-Based Feature Selection and Transformation (AAFST) mechanism, which leverages a Global Pooling Attention (GPA) strategy to identify and retain the most discriminative features for emotion classification.

The input to the AAFST mechanism is a tensor $\mathbf{X} \in \mathbb{R}^{N \times C \times E \times F}$, where N is the number of samples, $C = 62$ is the number of EEG channels, $E = 64$ is the number of epochs, and F is the number of features per channel.

2.3.1 Global Average Pooling

The first step summarizes temporal information across the epoch dimension via global average pooling:

$$\mathbf{P} = \frac{1}{E} \sum_{e=1}^{E} \mathbf{X}[:, :, e, :] \in \mathbb{R}^{N \times C \times F} \tag{10}$$

2.3.2 Attention Score Computation

Attention scores are computed by element-wise multiplication of the pooled features with a learnable weight matrix $\mathbf{W} \in \mathbb{R}^{C \times F}$, followed by a softmax normalization along the feature dimension:

$$\mathbf{A} = \text{softmax}(\mathbf{P} \odot \mathbf{W}) \in \mathbb{R}^{N \times C \times F} \tag{11}$$

where $\mathbf{W}$ is initialized randomly and optimized during training to capture channel-wise and feature-wise importance.

2.3.3 Feature Importance and Selection

Feature importance is derived by averaging the attention scores across all samples and channels:

$$\mathbf{I} = \frac{1}{N \cdot C} \sum_{n=1}^{N} \sum_{c=1}^{C} \mathbf{A}[n, c, :] \in \mathbb{R}^{F} \tag{12}$$

A dynamic threshold based on a percentile of the importance scores is then applied to construct a binary Selected Feature Mask (SFM):

$$\text{SFM} = \mathbf{I} > \tau \tag{13}$$

where τ is the percentile threshold. Only features whose importance exceeds τ are retained, effectively reducing the feature dimensionality while preserving the most emotionally discriminative information. The mask is applied to the pooled feature tensor to produce the final transformed representation, which is subsequently passed to the SVM classifier. The complete pipeline is presented as a pseudocode in Algorithm 1.

Algorithm 1 EEG-Based Emotion Recognition Pseudocode

```
Input: EEG Data (Channels × Samples)
Output: Emotion Classes
Algorithm:
Step 1: Load EEG Signals
eeg_data ← load_eeg_data()
Step 2: Preprocessing
downsampled ← downsample(eeg_data, 1000, 200)
filtered ← bandpass_filter(downsampled, 0.5, 75)
Step 3: Segmentation and Epoch Generation
epochs ← sliding_window(filtered, epoch_size=800, n_epochs=64)
Step 4: Channel-wise Feature Extraction
features ← [ ]
for each channel in epochs do
    channel_features ← extract_features(channel)
    features.append(channel_features)
end for
Step 5: AAFST using Global Pooling Attention
pooled ← (1/E) Σ_{e=1}^{E} X[:, :, e, :]
A ← softmax(pooled ⊙ W)
I ← mean(A, axis = (N, C))
τ ← percentile(I, p)
SFM ← I > τ
reduced_features ← features[:, :, SFM]
Step 6: Feature Flattening
flattened ← flatten(reduced_features)
Step 7: Multi-Class SVM Training
X_train, X_test, y_train, y_test ← train_test_split(flattened)
svm ← train_SVM(X_train, y_train, kernel=RBF, C=10)
Step 8: Prediction
predictions ← svm.predict(X_test)
Step 9: Evaluation
evaluate(predictions, y_test)
```

2.4 Support Vector Machine (SVM)

Support Vector Machines (SVM) are supervised learning models that identify the optimal hyperplane separating data points of different classes in a high-dimensional feature space. Given a set of training samples $\{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector and $y_i \in \{1, \ldots, K\}$ is the class label, the SVM seeks to maximize the margin between classes by solving the following optimization problem:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N} \xi_i \tag{14}$$

subject to:

$$y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{15}$$

where $\mathbf{w}$ is the weight vector, b is the bias term, C is the regularization parameter controlling the trade-off between margin maximization and classification error, and ξ_i are slack variables that allow for misclassification in non-linearly separable cases.

For non-linearly separable data, the Radial Basis Function (RBF) kernel is employed to implicitly map the input features into a higher-dimensional space:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{16}$$

where γ controls the influence of individual training samples. For multi-class classification across four emotional states, the one-vs-one strategy is adopted, constructing $\binom{4}{2} = 6$ binary classifiers and determining the final predicted class by majority voting. In this work, the SVM is trained with $C = 10$ and $\gamma = \text{scale}$, selected via grid search cross-validation.

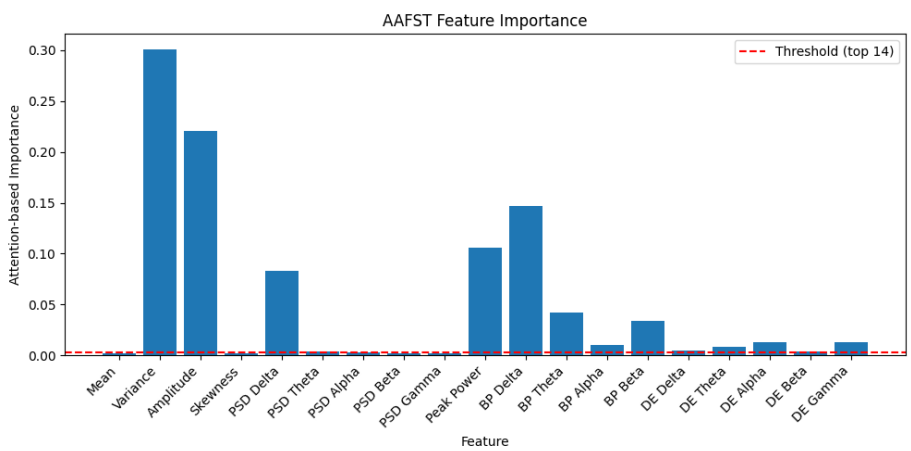

Figure 5: AAFST for Feature Importance

3 Results

Model	F1 Score (Weighted)
Vanilla SVM	53%
SVM with PCA	59%
SVM with AAFST	83%
SVM with AAFST + Windowed Epochs	85%

Table 1: Classification performance across model configurations.

We evaluate 4 variants of the SVM model. The baseline SVM with gridsearch trained on raw extracted features achieved a weighted F1 score of 53%,

Applying PCA for dimensionality reduction improved performance marginally to 59% by removing redundant features.

The introduction of the AAFST mechanism yielded a substantial improvement to 83%. The best performance of 85% was achieved by combining AAFST and forcing epochs to 64 by using overlapping windowed epochs, which significantly increased the number of training samples available to the classifier, improving generalization across the four emotional states.

```
Accuracy: 85.36%
              precision    recall  f1-score

     Neutral       0.82      0.87      0.85
    Positive       0.84      0.82      0.83
    Negative       0.86      0.85      0.86
       Mixed       0.89      0.87      0.88

    accuracy                           0.85
   macro avg       0.85      0.85      0.85
weighted avg       0.85      0.85      0.85
```

Figure 6: Result produced by best performing model SVM + AAFST + windowed epochs

4 Conclusion

In this work, we succefully implemented the methodology used in "Human Emotion Recognition from EEG Brain Signals using Enhanced Machine Learning Methods." Using the SEED-IV dataset, we demonstrated that preprocessing, handcrafted feature extraction, and attention-based feature selection collectively contribute to improved classification performance. The proposed pipeline

achieved a final weighted F1 score of 85% across four emotional states, closely replicating the results reported in the original paper. The AAFST mechanism proved to be the most impactful component, highlighting the importance of selective feature retention in high-dimensional EEG feature spaces.

Future work could explore deep learning-based end-to-end approaches and subject-independent cross-validation to further improve generalization.

4.1 Limitations

- No extensive fine-tuning was done beyond a basic grid search, so there is likely still room for improvement in classification performance.
- The AAFST mechanism was not fully implemented as described — ideally $\mathbf{W}$ should be optimized through a neural network, but we used a randomly initialized fixed matrix instead due to the complexity of the training setup.
- The paper was unclear about how the 64 epochs per trial were generated, so we assumed a sliding window approach with overlapping windows, which may or may not match what the authors actually did.
- The pipeline is quite slow to run — feature extraction across the full dataset takes a considerable amount of time, which made experimentation difficult.
- Deep learning approaches could be a promising future direction as they would eliminate the need for manual feature extraction altogether.
- The train/test split was done randomly, meaning epochs from the same trial can appear in both train and test sets. A proper subject-independent evaluation using leave-one-subject-out cross-validation would give a more realistic estimate of generalization performance.